



Yahya Ibrahim, Ph.D.

AI RESEARCHER · COMPUTER VISION
& 3D SCENE UNDERSTANDING

CONTACT

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RESEARCH IMPACT

11
PAPERS

163
CITATIONS

6
H-INDEX

5
I10-INDEX

TECHNICAL SKILLS

Programming

Python MATLAB C++

DL Frameworks

PyTorch TensorFlow Keras
Lightning

Architectures

Transformer Autoregressive
CNN GAN

3D & Vision

NeRF 3D Gaussian Splatting
OpenCV Open3D PCL

Tools & Infra

ROS2 Docker Git W&B
HPC/Linux LaTeX

EDUCATION

PhD · PPKE ITK, Budapest
2017 – 2023
Dissertation: "Deep Learning Based Methods on
2D Image and 3D Structure Inpainting"

MSc · Tishreen Univ., Syria
2013 – 2016

BSc · Tishreen Univ., Syria
2009 – 2013

LANGUAGES

Arabic (Native) English (Fluent)
Hungarian (Basic)

RESEARCH PROFILE

PhD graduate (PPKE ITK, 2017–2023) with six years of research at HUN-REN SZTAKI's Machine Perception Laboratory. Dissertation: "*Deep Learning Based Methods on 2D Image and 3D Structure Inpainting*". Research spans 3D scene reconstruction (point cloud completion, NeRF, 3D Gaussian Splatting), GAN-based generative image and structure inpainting, and satellite image analysis for geospatial AI — with 11 peer-reviewed publications, 163 citations, and h-index of 6. Seeking **Research Scientist** positions in computer vision and deep learning.

RESEARCH INTERESTS

3D Scene Reconstruction Point Cloud Completion NeRF & 3D Gaussian Splatting
GAN-Based Inpainting Remote Sensing & SAR Fusion Semantic Segmentation
Transformer Architectures Autoregressive 3D Models

RESEARCH EXPERIENCE

Researcher — Machine Perception Laboratory

2017 – Present

HUN-REN SZTAKI · Pázmány Péter Catholic University (PPKE ITK), Budapest

- Designed **MVPCC-Net**: multi-view deep learning architecture for 3D LiDAR point cloud completion in autonomous driving; published in Elsevier IVC (2023, 11 citations).
- Developed **GAN-based blind inpainting** pipeline for 2D masonry wall restoration and 3D structure completion without mask supervision (WACV 2021; MDPI Remote Sensing 2020, 39 citations).
- Built **SplitClass segmentation** framework fusing Sentinel-1 SAR and Sentinel-2 multispectral data with temporal gap-filling for inland flood monitoring (ICCVW 2025, IEEE/CVF).
- Contributed to **EEG-based BCI** via deep recurrent CNN for motor imagery classification (IEEE BCI 2020, 57 citations).
- Extending 3D perception to **drone and underwater LiDAR** sensing within the EU Proactif emergency management project (ROS2 pipeline).
- Research visit** to Nanyang Technological University (NTU) Singapore (Nov 2025) — S-Lab and CoNiC Lab collaboration on geospatial AI and generative multimedia.

PUBLICATIONS · scholar.google.com/citations?user=1ShbTvIAAAAJ

Inland Excess Water (IEW) Monitoring Using Sentinel-1/2: A SplitClass Segmentation and Temporal Gap-Filling Approach

Y. Ibrahim, M. Belényesi, C. Liu, M. Richter-Cserey, M. Simon, T. Sziranyi, C. Benedek
ICCVW — IEEE/CVF, SEA Workshop · 2025

MVPCC-Net: Multi-View Based Point Cloud Completion Network for MLS Data

Y. Ibrahim, C. Benedek
Image and Vision Computing — Elsevier · 2023 · 11 citations

Multi-View Based 3D Point Cloud Completion Algorithm for Vehicles in Autonomous Driving

Y. Ibrahim, B. Nagy, C. Benedek
ICPR — IEEE · 2022 · 7 citations

A GAN-Based Blind Inpainting Method for Masonry Wall Images

Y. Ibrahim, B. Nagy, C. Benedek
WACV — IEEE Winter Conf. on Applications of Computer Vision · 2021

Multi-Class Classification of Motor Imagery EEG Signals Using Deep Recurrent CNN

W. Fadel, C. Kollod, M. Wahdow, Y. Ibrahim, I. Ulbert
8th International BCI Meeting — IEEE · 2020 · 57 citations

Deep Learning-Based Masonry Wall Image Analysis

Y. Ibrahim, B. Nagy, C. Benedek
Remote Sensing — MDPI · 2020 · 39 citations

CNN-Based Watershed Marker Extraction for Brick Segmentation in Masonry Walls

Y. Ibrahim, B. Nagy, C. Benedek
ISVC — Springer International Symposium on Visual Computing · 2019

PROFESSIONAL SERVICE

Peer reviewer · 4 Elsevier Q1 journals · 9 reviews · ORCID verified (orcid.org/
0000-0003-0131-985X)

Engineering Applications of AI · 2025 Automation in Construction · 2024
Measurement · 2024